

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date: 26-02-26

School: SCOPE

Reg. no.: 23BCE20187

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and write a MongoDB query to select database as "Vehicles".

```
C:\Program Files\MongoDB\bin\mongo
MongoDB shell version v5.0.3
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("9957ada7-2d8a-47a8-9d93-bfabdec21887") }
MongoDB server version: 5.0.3
=====
Warning: the "mongo" shell has been superseded by "mongosh",
which delivers improved usability and compatibility. The "mongo" shell has been deprecated and will be removed in
an upcoming release.
We recommend you begin using "mongosh".
For installation instructions, see
https://docs.mongodb.com/mongodb-shell/install/
=====
The server generated these startup warnings when booting:
  2026-02-14T09:02:41.248+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----
  Enable MongoDB's free cloud-based monitoring service, which will then receive and display
  metrics about your deployment (disk utilization, CPU, operation statistics, etc).

  The monitoring data will be available on a MongoDB website with a unique URL accessible to you
  and anyone you share the URL with. MongoDB may use this information to make product
  improvements and to suggest MongoDB products and deployment options to you.

  To enable free monitoring, run the following command: db.enableFreeMonitoring()
  To permanently disable this reminder, run the following command: db.disableFreeMonitoring()
-----
> use vehicles
switched to db vehicles
> show dbs
admin            0.000GB
config           0.000GB
local            0.000GB
luffy            0.000GB
school           0.000GB
student          0.000GB
vehicles         0.000GB
> db.createCollection("two-wheelers", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("four-wheelers")
{ "ok" : 1 }
> db.two_wheels.insertMany([
...   {
...     bike_name: "Honda Activa 12",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["black", "silver", "blue", "white"],
...     manufacturer: "Honda",
...     performance: 8,
...     timestamp: ISODate("2023-01-15"),
...     price: 75000
...   },
...   {
...     bike_name: "TVS Apache RTR 160",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["red", "black", "blue", "white"],
...   }
... ])
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date: 26-02-26

School: SCOPE

Reg. no.: 23BCE20187

2. Write a MongoDB query to display all the databases.

```
C:\Program Files\MongoDB\bin\mongo X
MongoDB shell version v5.0.3
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("9957ada7-2d8a-47a8-9d93-bfabdec21887") }
MongoDB server version: 5.0.3

Warning: the "mongo" shell has been superseded by "mongosh",
which delivers improved usability and compatibility. The "mongo" shell has been deprecated and will be removed in
an upcoming release.
We recommend you begin using "mongosh".
For installation instructions, see
https://docs.mongodb.com/mongodb-shell/install/

-----
The server generated these startup warnings when booting:
  2026-02-14T09:02:41.248+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

  Enable MongoDB's free cloud-based monitoring service, which will then receive and display
  metrics about your deployment (disk utilization, CPU, operation statistics, etc).

  The monitoring data will be available on a MongoDB website with a unique URL accessible to you
  and anyone you share the URL with. MongoDB may use this information to make product
  improvements and to suggest MongoDB products and deployment options to you.

  To enable free monitoring, run the following command: db.enableFreeMonitoring()
  To permanently disable this reminder, run the following command: db.disableFreeMonitoring()
-----

> use vehicles
switched to db vehicles
> show dbs
admin            0.000GB
config           0.000GB
local            0.000GB
luffy            0.000GB
school           0.000GB
student          0.000GB
vehicles         0.000GB
> db.createCollection("two-wheelers", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("four-wheelers")
{ "ok" : 1 }
> db.two-wheelers.insertMany([
...   {
...     bike_name: "Honda Activa 12",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["black", "silver", "blue", "white"],
...     manufacturer: "Honda",
...     performance: 8,
...     timestamp: ISODate("2023-01-15"),
...     price: 75000
...   },
...   {
...     bike_name: "TVS Apache RTR 160",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["red", "black", "blue", "white"],
...   }
... ])
{ "ok" : 1 }
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date: 26-02-26

School: SCOPE

Reg. no.: 23BCE20187

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```

MongoDB shell version v5.0.3
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("9957ada7-2d0a-47a8-9d93-bfabdec21887") }
MongoDB server version: 5.0.3

Warning: the "mongo" shell has been superseded by "mongosh",
which delivers improved usability and compatibility. The "mongo" shell has been deprecated and will be removed in
an upcoming release.
We recommend you begin using "mongosh".
For installation instructions, see
https://docs.mongodb.com/mongodb-shell/install/

-----
The server generated these startup warnings when booting:
  2026-02-14T09:02:41.248+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

Enable MongoDB's free cloud-based monitoring service, which will then receive and display
metrics about your deployment (disk utilization, CPU, operation statistics, etc).

The monitoring data will be available on a MongoDB website with a unique URL accessible to you
and anyone you share the URL with. MongoDB may use this information to make product
improvements and to suggest MongoDB products and deployment options to you.

To enable free monitoring, run the following command: db.enableFreeMonitoring()
To permanently disable this reminder, run the following command: db.disableFreeMonitoring()
-----

> use vehicles
switched to db vehicles
> show dbs
admin            0.000GB
config           0.000GB
local            0.000GB
luffy            0.000GB
school           0.000GB
student          0.000GB
vehicles         0.000GB
> db.createCollection("two-wheelers", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("four-wheelers")
{ "ok" : 1 }
> db.two_wheelers.insertMany([
...   {
...     bike_name: "Honda Activa 12",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["black", "silver", "blue", "white"],
...     manufacturer: "Honda",
...     performance: 8,
...     timestamp: ISODate("2023-01-15"),
...     price: 75000
...   },
...   {
...     bike_name: "TVS Apache RTR 160B",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["red", "black", "blue", "white"],
...     manufacturer: "TVS",
...     performance: 9,
...     timestamp: ISODate("2022-06-20"),
...     price: 115000
...   },
...   {
...     bike_name: "Bajaj Pulsar NS200",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["sport red", "black", "blue", "orange"],
...     manufacturer: "Bajaj",
...     performance: 9,
...     timestamp: ISODate("2023-01-10"),
...     price: 110000
...   },
...   {
...     bike_name: "Suzuki Access 125",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["black", "grey", "red", "blue"],
...     manufacturer: "Suzuki",
...     performance: 8,
...     timestamp: ISODate("2022-11-05"),
...     price: 80000
...   },
...   {
...     bike_name: "Yamaha FZ-S",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["black", "blue", "red", "yellow"],
...     manufacturer: "Yamaha",
...     performance: 8,
...     timestamp: ISODate("2023-05-12"),
...     price: 120000
...   }
... ])
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0")
  ]
}

```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```

> db.two_wheelers.insertMany([
...   {
...     bike_name: "Honda Activa 12",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["black", "silver", "blue", "white"],
...     manufacturer: "Honda",
...     performance: 8,
...     timestamp: ISODate("2023-01-15"),
...     price: 75000
...   },
...   {
...     bike_name: "TVS Apache RTR 160B",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["red", "black", "blue", "white"],
...     manufacturer: "TVS",
...     performance: 9,
...     timestamp: ISODate("2022-06-20"),
...     price: 115000
...   },
...   {
...     bike_name: "Bajaj Pulsar NS200",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["sport red", "black", "blue", "orange"],
...     manufacturer: "Bajaj",
...     performance: 9,
...     timestamp: ISODate("2023-01-10"),
...     price: 110000
...   },
...   {
...     bike_name: "Suzuki Access 125",
...     model: "gearless",
...     category: "125cc",
...     colors_available: ["black", "grey", "red", "blue"],
...     manufacturer: "Suzuki",
...     performance: 8,
...     timestamp: ISODate("2022-11-05"),
...     price: 80000
...   },
...   {
...     bike_name: "Yamaha FZ-S",
...     model: "gear",
...     category: "160cc",
...     colors_available: ["black", "blue", "red", "yellow"],
...     manufacturer: "Yamaha",
...     performance: 8,
...     timestamp: ISODate("2023-05-12"),
...     price: 120000
...   }
... ])
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0"),
    ObjectId("699fc81580262fa080a0a0")
  ]
}

```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date: 26-02-26

School: SCOPE

Reg. no.: 23BCE20187

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
C:\Program Files\MongoDB\> db.createCollection("four-wheelers")
{ "ok" : 1 }
> db.two_wheelers.insertMany([
...
  {
    bike_name: "Honda Activa 12",
    model: "gearless",
    category: "125cc",
    colors_available: ["black", "silver", "blue", "white"],
    manufacturer: "Honda",
    performance: 8,
    timestamp: ISODate("2023-01-15"),
    price: 75000
  },
  {
    bike_name: "TVS Apache RTR 160",
    model: "gear",
    category: "160cc",
    colors_available: ["red", "black", "blue", "white"],
    manufacturer: "TVS",
    performance: 9,
    timestamp: ISODate("2022-06-20"),
    price: 115000
  },
  {
    bike_name: "Bajaj Pulsar NS200",
    model: "gear",
    category: "200cc",
    colors_available: ["sport red", "black", "blue", "orange"],
    manufacturer: "Bajaj",
    performance: 9,
    timestamp: ISODate("2023-03-10"),
    price: 135000
  },
  {
    bike_name: "Suzuki Access 125",
    model: "gearless",
    category: "125cc",
    colors_available: ["black", "grey", "red", "blue"],
    manufacturer: "Suzuki",
    performance: 7,
    timestamp: ISODate("2022-11-05"),
    price: 80000
  },
  {
    bike_name: "Yamaha FZ-S",
    model: "gear",
    category: "150cc",
    colors_available: ["black", "blue", "red", "yellow"],
    manufacturer: "Yamaha",
    performance: 8,
    timestamp: ISODate("2023-05-12"),
    price: 120000
  }
])
{
  "acknowledged" : true,
  "insertedIds" : [

```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:26-02-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Manne Keerthi
Sameera

Reg. no.: 23BCE20187

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
> db.two_wheelers.find()
{ "_id" : ObjectId("699fca1580202cfa9805a45"), "bike_name" : "Honda Active 12", "model" : "gearless", "category" : "125cc", "colors_available" : [ "black", "silver", "blue", "white" ], "manufacturer" : "Honda", "performance" : 8, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "price" : 75000 }
{ "_id" : ObjectId("699fca1580202cfa9805a46"), "bike_name" : "TVS Apache RTR 1600", "model" : "gear", "category" : "160cc", "colors_available" : [ "red", "black", "blue", "white" ], "manufacturer" : "TVS", "performance" : 9, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "price" : 115000 }
{ "_id" : ObjectId("699fca1580202cfa9805a47"), "bike_name" : "Bajaj Pulsar NS200", "model" : "gear", "category" : "200cc", "colors_available" : [ "sport red", "black", "blue", "orange" ], "manufacturer" : "Bajaj", "performance" : 9, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "price" : 135000 }
{ "_id" : ObjectId("699fca1580202cfa9805a48"), "bike_name" : "Suzuki Access 125", "model" : "gearless", "category" : "125cc", "colors_available" : [ "black", "grey", "red", "blue" ], "manufacturer" : "Suzuki", "performance" : 7, "timestamp" : ISODate("2022-11-05T00:00:00Z"), "price" : 80000 }
{ "_id" : ObjectId("699fca1580202cfa9805a49"), "bike_name" : "Yamaha FZ-5", "model" : "gear", "category" : "150cc", "colors_available" : [ "black", "blue", "red", "yellow" ], "manufacturer" : "Yamaha", "performance" : 8, "timestamp" : ISODate("2023-05-12T00:00:00Z"), "price" : 120000 }
> db.four_wheelers.insertMany([
...   {
...     vehicle_name: "Maruti Swift",
...     model: "own",
...     category: "car",
...     variants: ["LXI", "VXI", "ZXI", "petrol", "diesel"],
...     manufacturer: "Maruti Suzuki",
...     performance: 8,
...     timestamp: ISODate("2023-02-14"),
...     price: 600000
...   },
...   {
...     vehicle_name: "Tata Ace",
...     model: "commercial",
...     category: "mini truck",
...     variants: ["petrol", "diesel", "gold"],
...     manufacturer: "Tata",
...     performance: 7,
...     timestamp: ISODate("2022-08-22"),
...     price: 350000
...   },
...   {
...     vehicle_name: "Mahindra Bolero",
...     model: "commercial",
...     category: "SUV",
...     variants: ["LX", "ZX", "diesel"],
...     manufacturer: "Mahindra",
...     performance: 8,
...     timestamp: ISODate("2023-01-30"),
...     price: 900000
...   },
...   {
...     vehicle_name: "Eicher Pro 2095",
...     model: "commercial",
...     category: "lorry",
...     variants: ["H5", "fuel tanker", "container"],
...     manufacturer: "Eicher",
...     performance: 9,
...     timestamp: ISODate("2021-12-15"),
...     price: 1500000
...   },
...   {
...     vehicle_name: "Ashok Leyland Bus",
...     model: "commercial",
...     category: "bus",
...     variants: ["AC", "Non-AC", "sleeper", "sprinter"],
...     manufacturer: "Ashok Leyland",
...     performance: 9,
...     timestamp: ISODate("2022-05-10"),
...     price: 2500000
...   }
... ])
{ "acknowledged" : true,
  "insertedIds" : [
    ObjectId("699fcb09580202cfa9805a4a"),
    ObjectId("699fcb09580202cfa9805a4b"),
    ObjectId("699fcb09580202cfa9805a4c"),
  ]
}
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date:26-02-26

School: SCOPE

Reg. no.: 23BCE20187

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```

> db.four_wheelers.find()
{ "_id" : ObjectId("699fc0955802cfa9805a5a"), "vehicle_name" : "Maruti Swift", "model" : "om", "category" : "car", "variants" : [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer" : "Maruti Suzuki", "performance" : 8, "timestamp" : ISODate("2023-02-14T00:00:00Z"), "price" : 600000 }
{ "_id" : ObjectId("699fc0955802cfa9805a5b"), "vehicle_name" : "Tata Ace", "model" : "commercial", "category" : "mini truck", "variants" : [ "petrol", "diesel", "gold" ], "manufacturer" : "Tata", "performance" : 7, "timestamp" : ISODate("2022-08-22T00:00:00Z"), "price" : 350000 }
{ "_id" : ObjectId("699fc0955802cfa9805a5c"), "vehicle_name" : "Mahindra Bolero", "model" : "commercial", "category" : "SUV", "variants" : [ "LX", "ZX", "diesel" ], "manufacturer" : "Mahindra", "performance" : 8, "timestamp" : ISODate("2023-01-30T00:00:00Z"), "price" : 900000 }
{ "_id" : ObjectId("699fc0955802cfa9805a5d"), "vehicle_name" : "Eicher Pro 2095", "model" : "commercial", "category" : "lorry", "variants" : [ "HS", "fuel tanker", "container" ], "manufacturer" : "Eicher", "performance" : 9, "timestamp" : ISODate("2021-12-15T00:00:00Z"), "price" : 1500000 }
{ "_id" : ObjectId("699fc0955802cfa9805a5e"), "vehicle_name" : "Ashok Leyland Bus", "model" : "commercial", "category" : "bus", "variants" : [ "AC", "Non-AC", " sleeper", "seater" ], "manufacturer" : "Ashok Leyland", "performance" : 9, "timestamp" : ISODate("2022-05-10T00:00:00Z"), "price" : 2500000 }
> db.two_wheelers.find({ bike_name: 1, price: 1 })
{ "_id" : ObjectId("699fc0955802cfa9805a5f"), "bike_name" : "Honda Activa 11", "price" : 75000 }
{ "_id" : ObjectId("699fc0955802cfa9805a60"), "bike_name" : "TVS Apache RTR 1600", "price" : 115000 }
{ "_id" : ObjectId("699fc0955802cfa9805a61"), "bike_name" : "Bajaj Pulsar NS200", "price" : 115000 }
{ "_id" : ObjectId("699fc0955802cfa9805a62"), "bike_name" : "Suzuki Access 125", "price" : 80000 }
{ "_id" : ObjectId("699fc0955802cfa9805a63"), "bike_name" : "Yamaha FZ-5", "price" : 120000 }
> db.four_wheelers.find({ vehicle_name: 1, price: 1 })
{ "_id" : ObjectId("699fc0955802cfa9805a64"), "vehicle_name" : "Maruti Swift", "price" : 600000 }
{ "_id" : ObjectId("699fc0955802cfa9805a65"), "vehicle_name" : "Tata Ace", "price" : 350000 }
{ "_id" : ObjectId("699fc0955802cfa9805a66"), "vehicle_name" : "Mahindra Bolero", "price" : 900000 }
{ "_id" : ObjectId("699fc0955802cfa9805a67"), "vehicle_name" : "Eicher Pro 2095", "price" : 1500000 }
{ "_id" : ObjectId("699fc0955802cfa9805a68"), "vehicle_name" : "Ashok Leyland Bus", "price" : 2500000 }
> db.two_wheelers.find({ manufacturer: "Honda" })
{ "_id" : ObjectId("699fc0955802cfa9805a69"), "bike_name" : "Honda Activa 11", "model" : "gearless", "category" : "125cc", "colors_available" : [ "black", "silver", "blue", "white" ], "manufacturer" : "Honda", "performance" : 8, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "price" : 75000 }
> db.two_wheelers.find({ manufacturer: "TVS" })
{ "_id" : ObjectId("699fc0955802cfa9805a6a"), "bike_name" : "TVS Apache RTR 1600", "model" : "gear", "category" : "160cc", "colors_available" : [ "red", "black", "blue", "white" ], "manufacturer" : "TVS", "performance" : 9, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "price" : 115000 }
> db.two_wheelers.find({ manufacturer: "Bajaj" })
{ "_id" : ObjectId("699fc0955802cfa9805a6b"), "bike_name" : "Bajaj Pulsar NS200", "model" : "gear", "category" : "200cc", "colors_available" : [ "sport red", "black", "blue", "orange" ], "manufacturer" : "Bajaj", "performance" : 9, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "price" : 115000 }
> db.two_wheelers.find({ manufacturer: "Suzuki" })
{ "_id" : ObjectId("699fc0955802cfa9805a6c"), "bike_name" : "Suzuki Access 125", "model" : "gearless", "category" : "125cc", "colors_available" : [ "black", "grey", "red", "blue" ], "manufacturer" : "Suzuki", "performance" : 7, "timestamp" : ISODate("2022-11-05T00:00:00Z"), "price" : 80000 }
> db.two_wheelers.find({ manufacturer: "Yamaha" })
{ "_id" : ObjectId("699fc0955802cfa9805a6d"), "bike_name" : "Yamaha FZ-5", "model" : "gear", "category" : "150cc", "colors_available" : [ "black", "blue", "red", "yellow" ], "manufacturer" : "Yamaha", "performance" : 8, "timestamp" : ISODate("2023-05-12T00:00:00Z"), "price" : 120000 }
> db.four_wheelers.find({ variants: "diesel" })
{ "_id" : ObjectId("699fc0955802cfa9805a6e"), "vehicle_name" : "Maruti Swift", "model" : "om", "category" : "car", "variants" : [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer" : "Maruti Suzuki", "performance" : 8, "timestamp" : ISODate("2023-02-14T00:00:00Z"), "price" : 600000 }
{ "_id" : ObjectId("699fc0955802cfa9805a6f"), "vehicle_name" : "Tata Ace", "model" : "commercial", "category" : "mini truck", "variants" : [ "petrol", "diesel", "gold" ], "manufacturer" : "Tata", "performance" : 7, "timestamp" : ISODate("2022-08-22T00:00:00Z"), "price" : 350000 }
{ "_id" : ObjectId("699fc0955802cfa9805a70"), "vehicle_name" : "Mahindra Bolero", "model" : "commercial", "category" : "SUV", "variants" : [ "LX", "ZX", "diesel" ], "manufacturer" : "Mahindra", "performance" : 8, "timestamp" : ISODate("2023-01-30T00:00:00Z"), "price" : 900000 }
> db.two_wheelers.find(
...   { bike_name: 1, price: 1 },
...   { performance: 1, id: 0 }
... )
{ "bike_name" : "Honda Activa 11", "category" : "125cc", "manufacturer" : "Honda", "performance" : 8 }
{ "bike_name" : "TVS Apache RTR 1600", "category" : "160cc", "manufacturer" : "TVS", "performance" : 9 }
{ "bike_name" : "Bajaj Pulsar NS200", "category" : "200cc", "manufacturer" : "Bajaj", "performance" : 9 }
{ "bike_name" : "Suzuki Access 125", "category" : "125cc", "manufacturer" : "Suzuki", "performance" : 7 }
{ "bike_name" : "Yamaha FZ-5", "category" : "150cc", "manufacturer" : "Yamaha", "performance" : 8 }
```

Branch/ Class: B.Tech/M.Tech

Date:26-02-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Manne Keerthi
Sameera

Reg. no.: 23BCE20187

8. Write a MongoDB query to display two_wheelers from a particular company

```
{
  "_id": ObjectId("699fc8b958802cf9885a4a"), "vehicle_name": "Maruti Swift", "model": "omni", "category": "car", "variants": [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer": "Maruti Suzuki", "performance": 8, "timestamp": ISODate("2023-02-14T00:00:00Z"), "price": 600000 }
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "vehicle_name": "Tata Ace", "model": "commercial", "category": "mini truck", "variants": [ "petrol", "diesel", "gold" ], "manufacturer": "Tata", "performance": 7, "timestamp": ISODate("2022-08-22T00:00:00Z"), "price": 350000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "vehicle_name": "Mahindra Bolero", "model": "commercial", "category": "SUV", "variants": [ "LX", "ZX", "diesel" ], "manufacturer": "Mahindra", "performance": 8, "timestamp": ISODate("2023-01-30T00:00:00Z"), "price": 900000 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "vehicle_name": "Eicher Pro 2095", "model": "commercial", "category": "lorry", "variants": [ "HS", "fuel tanker", "container" ], "manufacturer": "Eicher", "performance": 9, "timestamp": ISODate("2021-12-15T00:00:00Z"), "price": 1500000 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "vehicle_name": "Ashok Leyland Bus", "model": "commercial", "category": "bus", "variants": [ "AC", "Non-AC", " sleeper", "seater" ], "manufacturer": "Ashok Leyland", "performance": 9, "timestamp": ISODate("2022-05-10T00:00:00Z"), "price": 2500000 }
> db.two_wheelers.find( { $or: [ { bike_name: 1, price: 1 } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "bike_name": "Honda Activa 12", "price": 75000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "bike_name": "TVS Apache RTR 160R", "price": 115000 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "bike_name": "Bajaj Pulsar NS200", "price": 135000 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "bike_name": "Suzuki Access 125", "price": 80000 }
{ "_id": ObjectId("699fc8b958802cf9885a4f"), "bike_name": "Yamaha FZ-5", "price": 120000 }
> db.two_wheelers.find( { $or: [ { bike_name: 1, price: 1 } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4a"), "vehicle_name": "Maruti Swift", "price": 600000 }
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "vehicle_name": "Tata Ace", "price": 350000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "vehicle_name": "Mahindra Bolero", "price": 900000 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "vehicle_name": "Eicher Pro 2095", "price": 1500000 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "vehicle_name": "Ashok Leyland Bus", "price": 2500000 }
> db.two_wheelers.find( { manufacturer: "Honda" } )
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "bike_name": "Honda Activa 12", "model": "gearless", "category": "125cc", "colors_available": [ "black", "silver", "blue", "white" ], "manufacturer": "Honda", "performance": 8, "timestamp": ISODate("2023-01-15T00:00:00Z"), "price": 75000 }
> db.two_wheelers.find( { manufacturer: "TVS" } )
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "bike_name": "TVS Apache RTR 160R", "model": "gear", "category": "160cc", "colors_available": [ "red", "black", "blue", "white" ], "manufacturer": "TVS", "performance": 9, "timestamp": ISODate("2022-06-20T00:00:00Z"), "price": 115000 }
> db.two_wheelers.find( { manufacturer: "Bajaj" } )
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "bike_name": "Bajaj Pulsar NS200", "model": "gear", "category": "200cc", "colors_available": [ "sport red", "black", "blue", "orange" ], "manufacturer": "Bajaj", "performance": 9, "timestamp": ISODate("2023-03-10T00:00:00Z"), "price": 135000 }
> db.two_wheelers.find( { manufacturer: "Suzuki" } )
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "bike_name": "Suzuki Access 125", "model": "gearless", "category": "125cc", "colors_available": [ "black", "grey", "red", "blue" ], "manufacturer": "Suzuki", "performance": 7, "timestamp": ISODate("2022-11-05T00:00:00Z"), "price": 80000 }
> db.two_wheelers.find( { manufacturer: "Yamaha" } )
{ "_id": ObjectId("699fc8b958802cf9885a4f"), "bike_name": "Yamaha FZ-5", "model": "gear", "category": "150cc", "colors_available": [ "black", "blue", "red", "yellow" ], "manufacturer": "Yamaha", "performance": 8, "timestamp": ISODate("2023-05-12T00:00:00Z"), "price": 120000 }
> db.two_wheelers.find( { $or: [ { variants: "diesel" } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4a"), "vehicle_name": "Maruti Swift", "model": "omni", "category": "car", "variants": [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer": "Maruti Suzuki", "performance": 8, "timestamp": ISODate("2023-02-14T00:00:00Z"), "price": 600000 }
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "vehicle_name": "Tata Ace", "model": "commercial", "category": "mini truck", "variants": [ "petrol", "diesel", "gold" ], "manufacturer": "Tata", "performance": 7, "timestamp": ISODate("2022-08-22T00:00:00Z"), "price": 350000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "vehicle_name": "Mahindra Bolero", "model": "commercial", "category": "SUV", "variants": [ "LX", "ZX", "diesel" ], "manufacturer": "Mahindra", "performance": 8, "timestamp": ISODate("2023-01-30T00:00:00Z"), "price": 900000 }
> db.two_wheelers.find( { $or: [ { performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, performance: 1, _id: 0 } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "bike_name": "Honda Activa 12", "category": "125cc", "manufacturer": "Honda", "performance": 8 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "bike_name": "TVS Apache RTR 160R", "category": "160cc", "manufacturer": "TVS", "performance": 9 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "bike_name": "Bajaj Pulsar NS200", "category": "200cc", "manufacturer": "Bajaj", "performance": 9 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "bike_name": "Suzuki Access 125", "category": "125cc", "manufacturer": "Suzuki", "performance": 7 }
{ "_id": ObjectId("699fc8b958802cf9885a4f"), "bike_name": "Yamaha FZ-5", "category": "150cc", "manufacturer": "Yamaha", "performance": 8 }
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
> db.four_wheelers.find()
{ "_id": ObjectId("699fc8b958802cf9885a4a"), "vehicle_name": "Maruti Swift", "model": "omni", "category": "car", "variants": [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer": "Maruti Suzuki", "performance": 8, "timestamp": ISODate("2023-02-14T00:00:00Z"), "price": 600000 }
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "vehicle_name": "Tata Ace", "model": "commercial", "category": "mini truck", "variants": [ "petrol", "diesel", "gold" ], "manufacturer": "Tata", "performance": 7, "timestamp": ISODate("2022-08-22T00:00:00Z"), "price": 350000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "vehicle_name": "Mahindra Bolero", "model": "commercial", "category": "SUV", "variants": [ "LX", "ZX", "diesel" ], "manufacturer": "Mahindra", "performance": 8, "timestamp": ISODate("2023-01-30T00:00:00Z"), "price": 900000 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "vehicle_name": "Eicher Pro 2095", "model": "commercial", "category": "lorry", "variants": [ "HS", "fuel tanker", "container" ], "manufacturer": "Eicher", "performance": 9, "timestamp": ISODate("2021-12-15T00:00:00Z"), "price": 1500000 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "vehicle_name": "Ashok Leyland Bus", "model": "commercial", "category": "bus", "variants": [ "AC", "Non-AC", " sleeper", "seater" ], "manufacturer": "Ashok Leyland", "performance": 9, "timestamp": ISODate("2022-05-10T00:00:00Z"), "price": 2500000 }
> db.two_wheelers.find( { $or: [ { bike_name: 1, price: 1 } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "bike_name": "Honda Activa 12", "price": 75000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "bike_name": "TVS Apache RTR 160R", "price": 115000 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "bike_name": "Bajaj Pulsar NS200", "price": 135000 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "bike_name": "Suzuki Access 125", "price": 80000 }
{ "_id": ObjectId("699fc8b958802cf9885a4f"), "bike_name": "Yamaha FZ-5", "price": 120000 }
> db.four_wheelers.find( { $or: [ { bike_name: 1, price: 1 } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4a"), "vehicle_name": "Maruti Swift", "price": 600000 }
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "vehicle_name": "Tata Ace", "price": 350000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "vehicle_name": "Mahindra Bolero", "price": 900000 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "vehicle_name": "Eicher Pro 2095", "price": 1500000 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "vehicle_name": "Ashok Leyland Bus", "price": 2500000 }
> db.two_wheelers.find( { manufacturer: "Honda" } )
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "bike_name": "Honda Activa 12", "model": "gearless", "category": "125cc", "colors_available": [ "black", "silver", "blue", "white" ], "manufacturer": "Honda", "performance": 8, "timestamp": ISODate("2023-01-15T00:00:00Z"), "price": 75000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "bike_name": "TVS Apache RTR 160R", "model": "gear", "category": "160cc", "colors_available": [ "red", "black", "blue", "white" ], "manufacturer": "TVS", "performance": 9, "timestamp": ISODate("2022-06-20T00:00:00Z"), "price": 115000 }
> db.two_wheelers.find( { manufacturer: "Bajaj" } )
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "bike_name": "Bajaj Pulsar NS200", "model": "gear", "category": "200cc", "colors_available": [ "sport red", "black", "blue", "orange" ], "manufacturer": "Bajaj", "performance": 9, "timestamp": ISODate("2023-03-10T00:00:00Z"), "price": 135000 }
> db.two_wheelers.find( { manufacturer: "Suzuki" } )
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "bike_name": "Suzuki Access 125", "model": "gearless", "category": "125cc", "colors_available": [ "black", "grey", "red", "blue" ], "manufacturer": "Suzuki", "performance": 7, "timestamp": ISODate("2022-11-05T00:00:00Z"), "price": 80000 }
> db.two_wheelers.find( { manufacturer: "Yamaha" } )
{ "_id": ObjectId("699fc8b958802cf9885a4f"), "bike_name": "Yamaha FZ-5", "model": "gear", "category": "150cc", "colors_available": [ "black", "blue", "red", "yellow" ], "manufacturer": "Yamaha", "performance": 8, "timestamp": ISODate("2023-05-12T00:00:00Z"), "price": 120000 }
> db.four_wheelers.find( { $or: [ { variants: "diesel" } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4a"), "vehicle_name": "Maruti Swift", "model": "omni", "category": "car", "variants": [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer": "Maruti Suzuki", "performance": 8, "timestamp": ISODate("2023-02-14T00:00:00Z"), "price": 600000 }
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "vehicle_name": "Tata Ace", "model": "commercial", "category": "mini truck", "variants": [ "petrol", "diesel", "gold" ], "manufacturer": "Tata", "performance": 7, "timestamp": ISODate("2022-08-22T00:00:00Z"), "price": 350000 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "vehicle_name": "Mahindra Bolero", "model": "commercial", "category": "SUV", "variants": [ "LX", "ZX", "diesel" ], "manufacturer": "Mahindra", "performance": 8, "timestamp": ISODate("2023-01-30T00:00:00Z"), "price": 900000 }
> db.two_wheelers.find( { $or: [ { performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, performance: 1, _id: 0 } ] } )
{ "_id": ObjectId("699fc8b958802cf9885a4b"), "bike_name": "Honda Activa 12", "category": "125cc", "manufacturer": "Honda", "performance": 8 }
{ "_id": ObjectId("699fc8b958802cf9885a4c"), "bike_name": "TVS Apache RTR 160R", "category": "160cc", "manufacturer": "TVS", "performance": 9 }
{ "_id": ObjectId("699fc8b958802cf9885a4d"), "bike_name": "Bajaj Pulsar NS200", "category": "200cc", "manufacturer": "Bajaj", "performance": 9 }
{ "_id": ObjectId("699fc8b958802cf9885a4e"), "bike_name": "Suzuki Access 125", "category": "125cc", "manufacturer": "Suzuki", "performance": 7 }
{ "_id": ObjectId("699fc8b958802cf9885a4f"), "bike_name": "Yamaha FZ-5", "category": "150cc", "manufacturer": "Yamaha", "performance": 8 }
```


Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:26-02-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Manne Keerthi
Sameera

Reg. no.: 23BCE20187

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```

> db.four_wheelers.find()
{ "_id" : ObjectId("699fc0955802cfa9805a4a"), "vehicle_name" : "Maruti Swift", "model" : "om", "category" : "car", "variants" : [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer" : "Maruti Suzuki", "performance" : 8, "timestamp" : ISODate("2023-02-14T00:00:00Z"), "price" : 600000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4b"), "vehicle_name" : "Tata Ace", "model" : "commercial", "category" : "mini truck", "variants" : [ "petrol", "diesel", "gold" ], "manufacturer" : "Tata", "performance" : 7, "timestamp" : ISODate("2022-08-22T00:00:00Z"), "price" : 350000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4c"), "vehicle_name" : "Mahindra Bolero", "model" : "commercial", "category" : "SUV", "variants" : [ "LX", "ZX", "diesel" ], "manufacturer" : "Mahindra", "performance" : 8, "timestamp" : ISODate("2023-01-30T00:00:00Z"), "price" : 900000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4d"), "vehicle_name" : "Eicher Pro 2095", "model" : "commercial", "category" : "lorry", "variants" : [ "HS", "fuel tanker", "container" ], "manufacturer" : "Eicher", "performance" : 9, "timestamp" : ISODate("2021-12-15T00:00:00Z"), "price" : 1500000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4e"), "vehicle_name" : "Ashok Leyland Bus", "model" : "commercial", "category" : "bus", "variants" : [ "AC", "Non-AC", " sleeper", "seater" ], "manufacturer" : "Ashok Leyland", "performance" : 9, "timestamp" : ISODate("2022-05-10T00:00:00Z"), "price" : 2500000 }
> db.two_wheelers.find({ $bike_name: 1, price: 1 })
{ "_id" : ObjectId("699fc0955802cfa9805a4b"), "bike_name" : "Honda Activa 11", "price" : 75000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4c"), "bike_name" : "TVS Apache RTR 160R", "price" : 115000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4d"), "bike_name" : "Bajaj Pulsar NS200", "price" : 135000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4e"), "bike_name" : "Suzuki Access 125", "price" : 80000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4f"), "bike_name" : "Yamaha FZ-5", "price" : 120000 }
> db.four_wheelers.find({ $vehicle_name: 1, price: 1 })
{ "_id" : ObjectId("699fc0955802cfa9805a4b"), "vehicle_name" : "Maruti Swift", "price" : 600000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4c"), "vehicle_name" : "Tata Ace", "price" : 350000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4d"), "vehicle_name" : "Mahindra Bolero", "price" : 900000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4e"), "vehicle_name" : "Eicher Pro 2095", "price" : 1500000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4f"), "vehicle_name" : "Ashok Leyland Bus", "price" : 2500000 }
> db.two_wheelers.find({ manufacturer: "Honda" })
{ "_id" : ObjectId("699fc0955802cfa9805a4b"), "bike_name" : "Honda Activa 11", "model" : "gearless", "category" : "125cc", "colors_available" : [ "black", "silver", "blue", "white" ], "manufacturer" : "Honda", "performance" : 8, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "price" : 75000 }
> db.two_wheelers.find({ manufacturer: "TVS" })
{ "_id" : ObjectId("699fc0955802cfa9805a4c"), "bike_name" : "TVS Apache RTR 160R", "model" : "gear", "category" : "160cc", "colors_available" : [ "red", "black", "blue", "white" ], "manufacturer" : "TVS", "performance" : 9, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "price" : 115000 }
> db.two_wheelers.find({ manufacturer: "Bajaj" })
{ "_id" : ObjectId("699fc0955802cfa9805a4d"), "bike_name" : "Bajaj Pulsar NS200", "model" : "gear", "category" : "200cc", "colors_available" : [ "sport red", "black", "blue", "orange" ], "manufacturer" : "Bajaj", "performance" : 9, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "price" : 135000 }
> db.two_wheelers.find({ manufacturer: "Suzuki" })
{ "_id" : ObjectId("699fc0955802cfa9805a4e"), "bike_name" : "Suzuki Access 125", "model" : "gearless", "category" : "125cc", "colors_available" : [ "black", "grey", "red", "blue" ], "manufacturer" : "Suzuki", "performance" : 7, "timestamp" : ISODate("2022-11-05T00:00:00Z"), "price" : 80000 }
> db.two_wheelers.find({ manufacturer: "Yamaha" })
{ "_id" : ObjectId("699fc0955802cfa9805a4f"), "bike_name" : "Yamaha FZ-5", "model" : "gear", "category" : "150cc", "colors_available" : [ "black", "blue", "red", "yellow" ], "manufacturer" : "Yamaha", "performance" : 8, "timestamp" : ISODate("2023-05-12T00:00:00Z"), "price" : 120000 }
> db.four_wheelers.find({ variants: "diesel" })
{ "_id" : ObjectId("699fc0955802cfa9805a4a"), "vehicle_name" : "Maruti Swift", "model" : "om", "category" : "car", "variants" : [ "LXI", "VXI", "ZXI", "petrol", "diesel" ], "manufacturer" : "Maruti Suzuki", "performance" : 8, "timestamp" : ISODate("2023-02-14T00:00:00Z"), "price" : 600000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4b"), "vehicle_name" : "Tata Ace", "model" : "commercial", "category" : "mini truck", "variants" : [ "petrol", "diesel", "gold" ], "manufacturer" : "Tata", "performance" : 7, "timestamp" : ISODate("2022-08-22T00:00:00Z"), "price" : 350000 }
{ "_id" : ObjectId("699fc0955802cfa9805a4c"), "vehicle_name" : "Mahindra Bolero", "model" : "commercial", "category" : "SUV", "variants" : [ "LX", "ZX", "diesel" ], "manufacturer" : "Mahindra", "performance" : 8, "timestamp" : ISODate("2023-01-30T00:00:00Z"), "price" : 900000 }
> db.two_wheelers.find(
... { performance: { $gt: 5 } },
... { bike_name: 1, category: 1, manufacturer: 1, performance: 1, _id: 0 }
... )
{ "bike_name" : "Honda Activa 11", "category" : "125cc", "manufacturer" : "Honda", "performance" : 8 }
{ "bike_name" : "TVS Apache RTR 160R", "category" : "160cc", "manufacturer" : "TVS", "performance" : 9 }
{ "bike_name" : "Bajaj Pulsar NS200", "category" : "200cc", "manufacturer" : "Bajaj", "performance" : 9 }
{ "bike_name" : "Suzuki Access 125", "category" : "125cc", "manufacturer" : "Suzuki", "performance" : 7 }
{ "bike_name" : "Yamaha FZ-5", "category" : "150cc", "manufacturer" : "Yamaha", "performance" : 8 }
}

```


Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date:26-02-26

School: SCOPE

Reg. no.: 23BCE20187

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and write a MongoDB query to select database as 'animal'.

```
C:\Program Files\MongoDB\bin> { "bike_name" : "Yamaha FZ-S", "category" : "150cc", "manufacturer" : "Yamaha", "performance" : 8 }
> use animals
switched to db animals
> db
animals
> show dbs
admin      0.000GB
config     0.000GB
local      0.000GB
luffy      0.000GB
school     0.000GB
student    0.000GB
vehicles   0.000GB
> db.createCollection("wild_animals", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
> db.wild_animals.insertMany([
...     {
...         animal_name: "Lion",
```

2. Write a MongoDB query to display all the databases.

```
C:\Program Files\MongoDB\bin> { "bike_name" : "Yamaha FZ-S", "category" : "150cc", "manufacturer" : "Yamaha", "performance" : 8 }
> use animals
switched to db animals
> db
animals
> show dbs
admin      0.000GB
config     0.000GB
local      0.000GB
luffy      0.000GB
school     0.000GB
student    0.000GB
vehicles   0.000GB
> db.createCollection("wild_animals", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
> db.wild_animals.insertMany([
...     {
...         animal_name: "Lion",
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:26-02-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Manne Keerthi
Sameera

Reg. no.: 23BCE20187

3. Create a collection called 'wild_animals' (use capping) and Create a collection called 'domestic_animals'.

```

C:\Program Files\MongoDB\bin\mongo
> use animals
switched to db animals
> db
animals
> show dbs
admin      0.000GB
config     0.000GB
local      0.000GB
luffy      0.000GB
school     0.000GB
student    0.000GB
vehicles   0.000GB
> db.createCollection("wild_animals", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
> db.wild_animals.insertMany([
...   {
...     animal_name: "Lion",

```

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```

> db.wild_animals.insertMany([
...   {
...     animal_name: "Lion",
...     nature: "harm",
...     favorite_foods: ["meat", "rabbits", "deer"],
...     care_taker_name: "John Smith",
...     life_span: 14,
...     timestamp: ISODate("2023-01-15"),
...     expenses: 50000
...   },
...   {
...     animal_name: "Elephant",
...     nature: "harmless",
...     favorite_foods: ["grass", "fruits", "bark", "roots"],
...     care_taker_name: "Rajesh Kumar",
...     life_span: 60,
...     timestamp: ISODate("2022-06-28"),
...     expenses: 80000
...   },
...   {
...     animal_name: "Tiger",
...     nature: "harm",
...     favorite_foods: ["meat", "deer", "wild boar"],
...     care_taker_name: "Mike Johnson",
...     life_span: 16,
...     timestamp: ISODate("2023-03-10"),
...     expenses: 35000
...   },
...   {
...     animal_name: "Giraffe",
...     nature: "harmless",
...     favorite_foods: ["leaves", "acacia", "fruits"],
...     care_taker_name: "Sarah Williams",
...     life_span: 25,
...     timestamp: ISODate("2022-11-05"),
...     expenses: 45000
...   },
...   {
...     animal_name: "Leopard",
...     nature: "harm",
...     favorite_foods: ["meat", "monkeys", "birds"],
...     care_taker_name: "David Brown",
...     life_span: 12,
...     timestamp: ISODate("2023-05-12"),
...     expenses: 40000
...   }
... ])
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("609fc9c580202cfa0805a1f"),
    ObjectId("609fc9c580202cfa0805a19"),
    ObjectId("609fc9c580202cfa0805a17"),
    ObjectId("609fc9c580202cfa0805a13"),
    ObjectId("609fc9c580202cfa0805a11")
  ]
}
> db.domestic_animals.insertMany([
...   {
...     animal_name: "Dog",

```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: Manne Keerthi
Sameera

Date:26-02-26

School: SCOPE

Reg. no.: 23BCE20187

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```

db.domestic_animals.insertMany([
  {
    animal_name: "Dog",
    gender: "male",
    favorite_foods: ["meat", "rice", "bones"],
    animal_petname: "Buddy",
    life_span: 12,
    timestamp: ISODate("2023-02-14"),
    expenses: 5000
  },
  {
    animal_name: "Cat",
    gender: "female",
    favorite_foods: ["fish", "milk", "meat"],
    animal_petname: "Whiskers",
    life_span: 15,
    timestamp: ISODate("2022-08-22"),
    expenses: 3000
  },
  {
    animal_name: "Cow",
    gender: "female",
    favorite_foods: ["grass", "hay", "grains"],
    animal_petname: "Gita",
    life_span: 20,
    timestamp: ISODate("2023-01-30"),
    expenses: 8000
  },
  {
    animal_name: "Horse",
    gender: "male",
    favorite_foods: ["grass", "oats", "carrots"],
    animal_petname: "Thunder",
    life_span: 25,
    timestamp: ISODate("2022-05-15"),
    expenses: 15000
  },
  {
    animal_name: "Goat",
    gender: "female",
    favorite_foods: ["grass", "leaves", "vegetables"],
    animal_petname: "Lily",
    life_span: 15,
    timestamp: ISODate("2023-04-10"),
    expenses: 2500
  }
])

"acknowledged" : true,
"insertedIds" : [
  ObjectId("699fcfae580202cfa9805a50"),
  ObjectId("699fcfae580202cfa9805a59"),
  ObjectId("699fcfae580202cfa9805a56"),
  ObjectId("699fcfae580202cfa9805a57"),
  ObjectId("699fcfae580202cfa9805a58")
]

db.wild_animals.find()
{ "_id" : ObjectId("699fc9c580202cfa9805a4f"), "animal_name" : "Lion", "nature" : "harm", "favorite_foods" : [ "meat", "zebras", "deer" ], "care_taker_name" : "John Smith", "life_span" : 14, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "expenses" : 50000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a50"), "animal_name" : "Elephant", "nature" : "harmless", "favorite_foods" : [ "grass", "fruits", "bark", "roots" ], "care_taker_name" : "Rajesh Kumar", "life_span" : 60, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "expenses" : 80000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a51"), "animal_name" : "Tiger", "nature" : "harm", "favorite_foods" : [ "meat", "deer", "wild boar" ], "care_taker_name" : "Mike Johnson", "life_span" : 16, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "expenses" : 55000 }

```

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.

```

db.wild_animals.find()
{ "_id" : ObjectId("699fc9c580202cfa9805a4f"), "animal_name" : "Lion", "nature" : "harm", "favorite_foods" : [ "meat", "zebras", "deer" ], "care_taker_name" : "John Smith", "life_span" : 14, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "expenses" : 50000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a50"), "animal_name" : "Elephant", "nature" : "harmless", "favorite_foods" : [ "grass", "fruits", "bark", "roots" ], "care_taker_name" : "Rajesh Kumar", "life_span" : 60, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "expenses" : 80000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a51"), "animal_name" : "Tiger", "nature" : "harm", "favorite_foods" : [ "meat", "deer", "wild boar" ], "care_taker_name" : "Mike Johnson", "life_span" : 16, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "expenses" : 55000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a52"), "animal_name" : "Leopard", "nature" : "harmless", "favorite_foods" : [ "leaves", "acacia", "fruits" ], "care_taker_name" : "Sarah Williams", "life_span" : 25, "timestamp" : ISODate("2022-11-05T00:00:00Z"), "expenses" : 45000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a53"), "animal_name" : "Leopard", "nature" : "harm", "favorite_foods" : [ "meat", "mosses", "birds" ], "care_taker_name" : "David Brown", "life_span" : 12, "timestamp" : ISODate("2023-05-17T00:00:00Z"), "expenses" : 40000 }

db.domestic_animals.find()
{ "_id" : ObjectId("699fcfae580202cfa9805a50"), "animal_name" : "Dog", "gender" : "male", "favorite_foods" : [ "meat", "rice", "bones" ], "animal_petname" : "Buddy", "life_span" : 12, "timestamp" : ISODate("2023-02-14T00:00:00Z"), "expenses" : 5000 }
{ "_id" : ObjectId("699fcfae580202cfa9805a59"), "animal_name" : "Cat", "gender" : "female", "favorite_foods" : [ "fish", "milk", "meat" ], "animal_petname" : "Whiskers", "life_span" : 15, "timestamp" : ISODate("2022-08-22T00:00:00Z"), "expenses" : 3000 }
{ "_id" : ObjectId("699fcfae580202cfa9805a56"), "animal_name" : "Cow", "gender" : "female", "favorite_foods" : [ "grass", "hay", "grains" ], "animal_petname" : "Gita", "life_span" : 20, "timestamp" : ISODate("2023-01-30T00:00:00Z"), "expenses" : 8000 }
{ "_id" : ObjectId("699fcfae580202cfa9805a57"), "animal_name" : "Horse", "gender" : "male", "favorite_foods" : [ "grass", "oats", "carrots" ], "animal_petname" : "Thunder", "life_span" : 25, "timestamp" : ISODate("2022-05-15T00:00:00Z"), "expenses" : 15000 }
{ "_id" : ObjectId("699fcfae580202cfa9805a58"), "animal_name" : "Goat", "gender" : "female", "favorite_foods" : [ "grass", "leaves", "vegetables" ], "animal_petname" : "Lily", "life_span" : 15, "timestamp" : ISODate("2023-04-10T00:00:00Z"), "expenses" : 2500 }

db.wild_animals.find( { animal_name : "Lion", "expenses" : 50000 } )
{ "_id" : ObjectId("699fc9c580202cfa9805a4f"), "animal_name" : "Lion", "nature" : "harm", "favorite_foods" : [ "meat", "zebras", "deer" ], "care_taker_name" : "John Smith", "life_span" : 14, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "expenses" : 50000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a50"), "animal_name" : "Elephant", "nature" : "harmless", "favorite_foods" : [ "grass", "fruits", "bark", "roots" ], "care_taker_name" : "Rajesh Kumar", "life_span" : 60, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "expenses" : 80000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a51"), "animal_name" : "Tiger", "nature" : "harm", "favorite_foods" : [ "meat", "deer", "wild boar" ], "care_taker_name" : "Mike Johnson", "life_span" : 16, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "expenses" : 55000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a52"), "animal_name" : "Leopard", "nature" : "harmless", "favorite_foods" : [ "leaves", "acacia", "fruits" ], "care_taker_name" : "Sarah Williams", "life_span" : 25, "timestamp" : ISODate("2022-11-05T00:00:00Z"), "expenses" : 45000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a53"), "animal_name" : "Leopard", "nature" : "harm", "favorite_foods" : [ "meat", "mosses", "birds" ], "care_taker_name" : "David Brown", "life_span" : 12, "timestamp" : ISODate("2023-05-17T00:00:00Z"), "expenses" : 40000 }

db.wild_animals.find( { animal_name : "Lion", "expenses" : 50000 } )
{ "_id" : ObjectId("699fc9c580202cfa9805a4f"), "animal_name" : "Lion", "nature" : "harm", "favorite_foods" : [ "meat", "zebras", "deer" ], "care_taker_name" : "John Smith", "life_span" : 14, "timestamp" : ISODate("2023-01-15T00:00:00Z"), "expenses" : 50000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a50"), "animal_name" : "Elephant", "nature" : "harmless", "favorite_foods" : [ "grass", "fruits", "bark", "roots" ], "care_taker_name" : "Rajesh Kumar", "life_span" : 60, "timestamp" : ISODate("2022-06-20T00:00:00Z"), "expenses" : 80000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a51"), "animal_name" : "Tiger", "nature" : "harm", "favorite_foods" : [ "meat", "deer", "wild boar" ], "care_taker_name" : "Mike Johnson", "life_span" : 16, "timestamp" : ISODate("2023-03-10T00:00:00Z"), "expenses" : 55000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a52"), "animal_name" : "Leopard", "nature" : "harmless", "favorite_foods" : [ "leaves", "acacia", "fruits" ], "care_taker_name" : "Sarah Williams", "life_span" : 25, "timestamp" : ISODate("2022-11-05T00:00:00Z"), "expenses" : 45000 }
{ "_id" : ObjectId("699fc9c580202cfa9805a53"), "animal_name" : "Leopard", "nature" : "harm", "favorite_foods" : [ "meat", "mosses", "birds" ], "care_taker_name" : "David Brown", "life_span" : 12, "timestamp" : ISODate("2023-05-17T00:00:00Z"), "expenses" : 40000 }

```

Sameera

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

[illegible]

8. Write a MongoDB query to display domestic_animals whose life is a particular year

[illegible]

9. Write a MongoDB query to display wild_animals available under a particular care taker

[illegible]

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:26-02-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Manne Keerthi
Sameera

Reg. no.: 23BCE20187

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```

db.wildAnimals.find()
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Lion", "nature": "harm", "favorite_foods": [ "meat", "straw", "beer" ], "care_taker_name": "John Smith", "life_span": 14, "timestamp": ISODate("2023-01-15T00:00:00Z"), "expenses": 50000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Elephant", "nature": "harmless", "favorite_foods": [ "grass", "fruits", "honey", "root" ], "care_taker_name": "Rajesh Kumar", "life_span": 40, "timestamp": ISODate("2022-06-20T00:00:00Z"), "expenses": 80000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Tiger", "nature": "harm", "favorite_foods": [ "meat", "deer", "wild bear", "deer", "tiger", "tiger" ], "care_taker_name": "Mike Johnson", "life_span": 15, "timestamp": ISODate("2022-03-10T00:00:00Z"), "expenses": 30000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Gorilla", "nature": "harmless", "favorite_foods": [ "fruits", "vegetables", "nuts" ], "care_taker_name": "Sarah Williams", "life_span": 20, "timestamp": ISODate("2022-11-05T00:00:00Z"), "expenses": 40000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Leopard", "nature": "harm", "favorite_foods": [ "meat", "honey", "birds" ], "care_taker_name": "David Brown", "life_span": 12, "timestamp": ISODate("2023-05-21T00:00:00Z"), "expenses": 40000 }
}

db.wildAnimals.find( {
  "life_span": {
    "$gt": 5
  }
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Dog", "gender": "male", "favorite_foods": [ "meat", "rice", "honey" ], "animal_petname": "Buddy", "life_span": 12, "timestamp": ISODate("2023-02-10T00:00:00Z"), "expenses": 5000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Cat", "gender": "female", "favorite_foods": [ "fish", "milk", "meat" ], "animal_petname": "Mittens", "life_span": 15, "timestamp": ISODate("2022-08-22T00:00:00Z"), "expenses": 3000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Cow", "gender": "male", "favorite_foods": [ "grass", "hay", "grain" ], "animal_petname": "Gita", "life_span": 20, "timestamp": ISODate("2023-01-30T00:00:00Z"), "expenses": 8000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Horse", "gender": "male", "favorite_foods": [ "grass", "milk", "honey" ], "animal_petname": "Hobbes", "life_span": 25, "timestamp": ISODate("2022-07-10T00:00:00Z"), "expenses": 10000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Goat", "gender": "female", "favorite_foods": [ "grass", "leaves", "vegetables" ], "animal_petname": "Lily", "life_span": 13, "timestamp": ISODate("2023-06-10T00:00:00Z"), "expenses": 2000 }
}

db.wildAnimals.find( {
  "life_span": {
    "$gt": 10
  }
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Lion", "expenses": 50000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Elephant", "expenses": 80000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Tiger", "expenses": 30000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Gorilla", "expenses": 40000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Leopard", "expenses": 40000 }
}

db.wildAnimals.find( {
  "life_span": {
    "$gt": 12
  }
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Dog", "gender": "male", "favorite_foods": [ "meat", "rice", "honey" ], "animal_petname": "Buddy", "life_span": 12, "timestamp": ISODate("2023-02-10T00:00:00Z"), "expenses": 5000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Cat", "gender": "female", "favorite_foods": [ "fish", "milk", "meat" ], "animal_petname": "Mittens", "life_span": 15, "timestamp": ISODate("2022-08-22T00:00:00Z"), "expenses": 3000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Cow", "gender": "male", "favorite_foods": [ "grass", "hay", "grain" ], "animal_petname": "Gita", "life_span": 20, "timestamp": ISODate("2023-01-30T00:00:00Z"), "expenses": 8000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Horse", "gender": "male", "favorite_foods": [ "grass", "milk", "honey" ], "animal_petname": "Hobbes", "life_span": 25, "timestamp": ISODate("2022-07-10T00:00:00Z"), "expenses": 10000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Goat", "gender": "female", "favorite_foods": [ "grass", "leaves", "vegetables" ], "animal_petname": "Lily", "life_span": 13, "timestamp": ISODate("2023-06-10T00:00:00Z"), "expenses": 2000 }
}

db.wildAnimals.find( {
  "care_taker_name": "John Smith"
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Lion", "nature": "harm", "favorite_foods": [ "meat", "straw", "beer" ], "care_taker_name": "John Smith", "life_span": 14, "timestamp": ISODate("2023-01-15T00:00:00Z"), "expenses": 50000 }
}

db.wildAnimals.find( {
  "care_taker_name": "Rajesh Kumar"
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Elephant", "nature": "harmless", "favorite_foods": [ "grass", "fruits", "honey", "root" ], "care_taker_name": "Rajesh Kumar", "life_span": 40, "timestamp": ISODate("2022-06-20T00:00:00Z"), "expenses": 80000 }
}

db.wildAnimals.find( {
  "care_taker_name": "Mike Johnson"
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Tiger", "nature": "harm", "favorite_foods": [ "meat", "deer", "wild bear", "deer", "tiger", "tiger" ], "care_taker_name": "Mike Johnson", "life_span": 15, "timestamp": ISODate("2022-03-10T00:00:00Z"), "expenses": 30000 }
}

db.wildAnimals.find( {
  "care_taker_name": "Sarah Williams"
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Gorilla", "nature": "harmless", "favorite_foods": [ "fruits", "vegetables", "nuts" ], "care_taker_name": "Sarah Williams", "life_span": 20, "timestamp": ISODate("2022-11-05T00:00:00Z"), "expenses": 40000 }
}

db.wildAnimals.find( {
  "care_taker_name": "David Brown"
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Leopard", "nature": "harm", "favorite_foods": [ "meat", "honey", "birds" ], "care_taker_name": "David Brown", "life_span": 12, "timestamp": ISODate("2023-05-21T00:00:00Z"), "expenses": 40000 }
}

db.wildAnimals.find( {
  "life_span": {
    "$gt": 5
  },
  "expenses": {
    "$gt": 1
  },
  "_id": 0
} )
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Lion", "favorite_foods": [ "meat", "straw", "beer" ], "expenses": 50000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Elephant", "favorite_foods": [ "grass", "fruits", "honey", "root" ], "expenses": 80000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Tiger", "favorite_foods": [ "meat", "deer", "wild bear", "deer", "tiger", "tiger" ], "expenses": 30000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Gorilla", "favorite_foods": [ "fruits", "vegetables", "nuts" ], "expenses": 40000 }
{
  "_id": "6096c4c8032c4f885a2e", "animal_name": "Leopard", "favorite_foods": [ "meat", "honey", "birds" ], "expenses": 40000 }
}

```